

GRIEVANCE MANAGEMENT SYSTEM

Common Project Documentation

Prepared For: Academic / Professional Project Submission

Technology Stack

Frontend, Backend, Database, and Deployment Tools

Architecture & Project Type

Web-based Application | Modular System Architecture

TABLE OF CONTENTS

1. Project Goals & Objectives
 - a. Overview
 - b. Goals
 - c. Objectives
2. Scope of the Project
3. Project Timeline
4. Database Overview
5. Functional Requirements
6. Screen List
7. Screen Design Reference
8. Technology Stack
9. Project Structure
10. Professional Summary



Project Goals & Objectives

Overview

The Grievance Management System is a full-stack web-based application developed using ASP.NET Core MVC.

It provides a structured and transparent mechanism for grievance submission, assignment, review, tracking, escalation, and closure.

The system supports role-based access for Admin, Grievance Officer, Department Head, and Employee/Student/User.

It is designed using MVC architecture following modern software engineering practices.

The primary goal is to improve efficiency, accountability, transparency, and response time in grievance handling.

Goals

1. Design a Structured and Normalized Database
2. Implement Complete Grievance Lifecycle Management
3. Develop Secure Authentication and Authorization
4. Ensure Transparency and Traceability
5. Provide a Professional and User-Friendly Interface
6. Encourage Scalable and Maintainable Development Practices
7. Support Monitoring and Decision-Making

Objectives

1. User Management
2. Role and Access Management
3. Department Management
4. Grievance Category Management
5. Grievance Submission
6. Grievance Assignment and Processing
7. Status Tracking
8. Escalation Management
9. Resolution and Closure
10. Notifications and Communication
11. Reporting and Dashboard
12. Data Validation and Security

Scope of the Project

1. User and Role Management

Manage users, roles, profile information, and access permissions with role-based login and secure module access.

2. Department and Category Configuration

Manage master data for departments, grievance categories, priority levels, and statuses with dynamic configuration.

3. Grievance Registration

Allow end users to lodge grievances online with detailed descriptions and optional file attachments, generate acknowledgment details and reference numbers.

4. Grievance Review and Assignment

Enable responsible authorities to review incoming grievances and assign them to relevant department or officer, maintain assignment history.

5. Grievance Processing Workflow

Support status changes, internal remarks, action logs, reassignments, escalations, and resolution updates with traceable workflow.

6. Escalation and SLA Monitoring

Identify grievances that remain unresolved beyond expected timelines and escalate them automatically or manually.

7. Search, Filter, and Tracking

Provide advanced search and filter options based on grievance number, date range, category, department, priority, status, and assigned officer.

8. Reporting and Analytics

Generate management reports to evaluate department responsiveness, grievance trends, and resolution efficiency with printable and exportable features.

9. Audit and Compliance Support

Maintain audit logs for key system actions and preserve historical data for review, accountability, and analysis.

10. Responsive Full-Stack MVC Application

Deliver a complete web-based solution using ASP.NET Core MVC, Entity Framework Core, SQL Server, Razor Views, Bootstrap, and JavaScript.

Project Timeline

Week No.	Week Date	Task List
Week 1	22/06/2026 To 27/06/2026	Requirement gathering, domain understanding, and study of existing grievance handling systems.
Week 2	29/06/2026 To 04/07/2026	Preparation of software requirement documentation, module identification, and workflow definition.
Week 3	06/07/2026 To 11/07/2026	UI wireframing, layout design, navigation planning, and screen list finalization.
Week 4	13/07/2026 To 18/07/2026	ASP.NET Core MVC project setup, solution structuring, and database design finalization.
Week 5	20/07/2026 To 25/07/2026	Development of master modules such as Roles, Users, Departments, Categories, Status, and Priority.
Week 6	27/07/2026 To 01/08/2026	Development of grievance registration, grievance listing, and grievance detail modules.
Week 7	03/08/2026 To 08/08/2026	Development of grievance assignment, action remarks, and status transition modules.
Week 8	10/08/2026 To 15/08/2026	Development of escalation logic, resolution process, closure workflow, and attachment handling.
Week 9	17/08/2026 To 22/08/2026	Dashboard development, summary statistics, filtering, and role-based home screens.
Week 10	24/08/2026 To 29/08/2026	Form validations, server-side business rules, exception handling, and user-friendly messaging.
Week 11	31/08/2026 To 05/09/2026	Authentication, authorization, session handling, and secure access control implementation.
Week 12	07/09/2026 To 12/09/2026	Reporting module, export support, print-ready views, and audit tracking implementation.
Week 13	14/09/2026 To 19/09/2026	Testing, bug fixing, performance refinement, and usability improvements.
Week 14	21/09/2026 To 26/09/2026	Final review, documentation completion, deployment preparation, and project evaluation.

Database Overview

Major Tables

Table Name	Purpose
GMSRole	Stores system roles
GMSUser	Stores user account information
GMSUserRole	Maps users to roles
GMSDepartment	Stores department details
GMSCategory	Stores grievance categories
GMSPriority	Stores grievance priority definitions
GMSStatus	Stores grievance status values
GMSGrievance	Stores main grievance records
GMSGrievanceAssignment	Stores grievance assignment history
GMSGrievanceRemark	Stores remarks and action notes
GMSGrievanceAttachment	Stores document references
GMSEscalation	Stores escalation details
GMSAuditLog	Stores audit trail of important actions

GMSRole Table Schema

ColumnName	DataType	Remarks
RoleId	int	Primary Key
RoleName	nvarchar(50)	Not Null, Unique
Description	nvarchar(250)	Allow Null
IsActive	bit	Not Null

GMSUser Table Schema

ColumnName	DataType	Remarks
UserId	int	Primary Key
FullName	nvarchar(150)	Not Null
Email	nvarchar(150)	Not Null, Unique
PasswordHash	nvarchar(max)	Not Null
MobileNumber	nvarchar(15)	Not Null
DepartmentId	int	FK -> Department
Designation	nvarchar(100)	Allow Null
ProfilePicturePath	nvarchar(500)	Allow Null
IsActive	bit	Not Null
IsDeleted	bit	Allow Null, Soft Delete
CreatedAt	datetime	Not Null
UpdatedAt	datetime	Allow Null

Supporting Table Schemas

GMSUserRole Table Schema

ColumnName	DataType	Remarks
UserRoleId	int	Primary Key
UserId	int	Not Null, FK -> User
RoleId	int	Not Null, FK -> Role

GMSDepartment Table Schema

ColumnName	DataType	Remarks
DepartmentId	int	Primary Key
DepartmentName	nvarchar(100)	Not Null, Unique
Description	nvarchar(250)	Allow Null
IsActive	bit	Not Null

GMSCategory Table Schema

ColumnName	DataType	Remarks
CategoryId	int	Primary Key
CategoryName	nvarchar(100)	Not Null, Unique
Description	nvarchar(250)	Allow Null
IsActive	bit	Not Null

Priority and Status Table Schemas

Create database schema cards for three more tables:

GMSPriority Table Schema

ColumnName	DataType	Remarks
PriorityId	int	Primary Key
PriorityName	nvarchar(20)	Not Null
PriorityCssClass	nvarchar(50)	Not Null

GMSStatus Table Schema

ColumnName	DataType	Remarks
StatusId	int	Primary Key
StatusName	nvarchar(30)	Not Null
StatusCssClass	nvarchar(50)	Not Null

GMSGrievance Table Schema (Main Grievance Records)

ColumnName	DataType	Remarks
GrievanceId	int	Primary Key
GrievanceNumber	nvarchar(30)	Not Null, Unique
Title	nvarchar(200)	Not Null
Description	nvarchar(max)	Not Null
CategoryId	int	Not Null, FK -> Category
DepartmentId	int	Not Null, FK -> Department
SubmittedByUserId	int	Not Null, FK -> User
AssignedToUserId	int	Allow Null, FK -> User
PriorityId	int	Not Null, FK -> Priority
StatusId	int	Not Null, FK -> Status
SubmittedDate	datetime	Not Null, Default GETDATE()
DueDate	datetime	Allow Null
ResolvedDate	datetime	Allow Null
ClosedDate	datetime	Allow Null
ResolutionSummary	nvarchar(1000)	Allow Null
IsEscalated	bit	Not Null
IsDeleted	bit	Allow Null, Soft Delete

Grievance Transaction Tables

GMSGrievanceAssignment Table Schema

ColumnName	DataType	Remarks
AssignmentId	int	Primary Key
GrievanceId	int	Not Null, FK -> Grievance
AssignedFromUserId	int	Not Null, FK -> User
AssignedToUserId	int	Not Null, FK -> User
AssignedDate	datetime	Not Null
DueDate	datetime	Allow Null
AssignmentRemark	nvarchar(500)	Allow Null

GMSGrievanceRemark Table Schema

ColumnName	DataType	Remarks
RemarkId	int	Primary Key
GrievanceId	int	Not Null, FK -> Grievance
RemarkByUserId	int	Not Null, FK -> User
RemarkText	nvarchar(1000)	Not Null
StatusId	int	Allow Null, FK -> Status
CreatedAt	datetime	Not Null

GMSGrievanceAttachment Table Schema

ColumnName	DataType	Remarks
AttachmentId	int	Primary Key
GrievanceId	int	Not Null, FK -> Grievance
FileName	nvarchar(255)	Not Null
FilePath	nvarchar(500)	Not Null
UploadedByUserId	int	Not Null, FK -> User
UploadedAt	datetime	Not Null

Escalation and Audit Tables

GMSEscalation Table Schema

ColumnName	DataType	Remarks
EscalationId	int	Primary Key
GrievanceId	int	Not Null, FK -> Grievance
EscalatedFromUserId	int	Not Null, FK -> User
EscalatedToUserId	int	Not Null, FK -> User
EscalationLevel	int	Not Null
Reason	nvarchar(500)	Not Null
EscalationDate	datetime	Not Null

GMSAuditLog Table Schema

ColumnName	DataType	Remarks
AuditLogId	int	Primary Key
TableName	nvarchar(100)	Not Null
RecordId	int	Not Null
ActionType	nvarchar(50)	Not Null
ActionByUserId	int	Not Null, FK -> User
ActionDate	datetime	Not Null
OldValue	nvarchar(max)	Allow Null
NewValue	nvarchar(max)	Allow Null

Functional Requirements

1. Role Management

Priority: 1

- Manage roles such as Admin, Grievance Officer, Department Head, and User
- Prevent duplicate role definitions
- Enable activation and deactivation of roles

2. User Management

Priority: 1

- Manage create, view, edit, and deactivate operations for users
- Prevent duplicate email registrations
- Maintain user profile and department association

3. Department Management

Priority: 1

- Manage department master records
- Prevent duplicate department names
- Associate departments with responsible users or officers

4. Category Management

Priority: 2

- Manage grievance categories and subcategories
- Prevent duplicate category entries
- Support categorization for routing and reporting

5. Grievance Submission

Priority: 2

- Register grievances with title, description, department, category, priority, and attachments
- Generate a unique grievance number automatically
- Acknowledge successful grievance registration

6. Grievance Assignment

Priority: 2

- Assign grievances to officers or departments
- Capture assignment date, due date, and remarks
- Maintain reassignment history

7. Grievance Processing

Priority: 3

- Update grievance status throughout the processing lifecycle
- Add remarks, actions taken, and recommendations
- Track all internal updates against the grievance record

8. Escalation Management

Priority: 3

- Escalate unresolved or overdue grievances
- Maintain escalation levels and escalation reasons
- Notify relevant higher authorities where applicable

9. Resolution and Closure

Priority: 3

- Mark grievances as resolved with resolution remarks
- Allow closure after review or confirmation
- Maintain final resolution date and closure record

10. Dashboard and Reporting

Priority: 4

- Provide dashboard summaries based on user roles
- Show pending, resolved, escalated, overdue, and department-wise counts
- Generate reports using filters and date ranges

11. Authentication and Authorization

Priority: 4

- Implement secure login and logout mechanisms
- Restrict access to pages and actions based on assigned roles
- Prevent unauthorized access to protected data and workflows

12. Audit and Activity Tracking

Priority: 4

- Record critical actions such as grievance creation, assignment, status updates, escalation, and closure
- Support accountability and administrative review

Screen List

AUTHENTICATION AND USER ACCESS

- **Login Page:** Role-based login for Admin, Grievance Officer, Department Head, and User
- **Forgot Password / Reset Password:** Password recovery and reset process for authorized users
- **Profile Page:** View and update personal details, contact information, and profile image

MASTER MODULES

Role Management

- Role List Page
- Add/Edit Role Page

User Management

- User List Page
- Add/Edit User Page
- User Detail/Profile Page

Department Management

- Department List Page
- Add/Edit Department Page

Category Management

- Category List Page
- Add/Edit Category Page

Priority Management

- Priority List Page
- Add/Edit Priority Page

Status Management

- Status List Page
- Add/Edit Status Page

GRIEVANCE MANAGEMENT MODULES

Submit Grievance

- Grievance Registration Form
- Attachment Upload

Manage Grievances

- Grievance List Page
- Grievance Detail View
- Edit Grievance Page

Assignment Management

- Assign Grievance Page
- Reassign Grievance Page
- Assignment History View

Remarks and Action Tracking

- Add Remark Page
- Action History Timeline

Escalation Management

- Escalate Grievance Page
- Escalation History Page

Resolution and Closure

- Resolution Update Page
- Closure Confirmation Page

DASHBOARDS AND REPORTS

Admin Dashboard

- Total grievances
- Pending grievances
- Resolved grievances
- Escalated grievances
- Department-wise grievance summary

Grievance Officer Dashboard

- Assigned grievances
- In-progress cases
- Overdue cases
- Recently updated grievances

Department Head Dashboard

- Department grievances
- Resolved vs pending department cases
- Escalated departmental grievances

User Dashboard

- Submitted grievances
- Open grievances
- Resolved grievances
- Recent grievance updates

Reports

- Date-wise grievance report
- Department-wise report
- Category-wise report

Screen Design Reference

LOGIN PAGE

- Username or email input field
- Password input field
- Remember me checkbox option
- Forgot password link
- Secure login button
- Professional branding and layout

DASHBOARD

- Role-based summary cards showing key metrics
- Charts for grievance distribution and trends
- Recent activity section with latest updates
- Pending actions panel for quick access
- Quick access shortcuts to frequently used modules

MANAGE ROLES

- Role listing with search and pagination functionality
- Add, edit, activate, and deactivate action buttons
- Role description and status display

MANAGE USERS

- User listing with department and role filter options
- Add/edit form with comprehensive validation
- User profile detail section with contact information
- Status indicators for active/inactive users

MANAGE DEPARTMENTS

- Department list with search functionality
- Department configuration form
- Department-wise officer mapping section
- Active/inactive status management

MANAGE CATEGORIES

- Category and subcategory list view
- Add/edit configuration form
- Category description and status display
- Hierarchical category structure support

GRIEVANCE LIST PAGE

- Search by grievance number functionality
- Filter options by status, category, department, priority, and date range
- View, edit, assign, escalate, and resolve action buttons
- Pagination and sorting capabilities
- Status indicators with color coding

GRIEVANCE REGISTRATION PAGE

- Title input field
- Description textarea for detailed information
- Category dropdown selection
- Department dropdown selection
- Priority level selection
- Attachment upload functionality
- Submit and reset buttons
- Form validation messages

GRIEVANCE DETAIL PAGE

- Complete grievance information display
- Status timeline showing progression
- Assignment details with assigned officer information
- Remarks history with timestamps
- Attachment section with download options
- Escalation and resolution records
- Action buttons for status updates

ASSIGNMENT PAGE

- Assign to user/department dropdown
- Due date picker
- Assignment remark textarea
- Save and notify action button
- Validation for required fields

RESOLUTION PAGE

- Resolution summary textarea
- Supporting remark input field
- Status update dropdown
- Close grievance checkbox option
- Save and confirm button

REPORTS PAGE

- Comprehensive filters and search options
- Tabular data display with sorting
- Export to Excel/PDF options
- Print-ready view functionality
- Summary statistics and metrics
- Date range selection for reports

Suggested Technology Stack

Layer	Technology
Frontend	ASP.NET Core MVC, Razor Views, Bootstrap, JavaScript, jQuery
Backend	ASP.NET Core MVC, C#
ORM	Entity Framework Core
Database	SQL Server
Authentication	ASP.NET Core Identity / Cookie Authentication
Reporting	RDLC / Export to Excel/PDF
Architecture	MVC + Layered Architecture
Version Control	Git / GitHub

Suggested MVC Project Structure

```
GrievanceManagementSystem/
├── Controllers/
│   ├── AccountController.cs
│   ├── DashboardController.cs
│   ├── GrievanceController.cs
│   ├── UserController.cs
│   ├── RoleController.cs
│   ├── DepartmentController.cs
│   ├── CategoryController.cs
│   └── ReportController.cs
├── Models/
│   ├── GMSRole.cs
│   ├── GMSUser.cs
│   ├── GMSDepartment.cs
│   ├── GMSCategory.cs
│   ├── MSGGrievance.cs
│   ├── MSGGrievanceAssignment.cs
│   ├── MSGGrievanceRemark.cs
│   ├── MSGGrievanceAttachment.cs
│   ├── GMSEscalation.cs
│   └── GMSAuditLog.cs
├── ViewModels/
│   ├── GrievanceViewModel.cs
│   ├── UserViewModel.cs
│   ├── DashboardViewModel.cs
│   └── ReportViewModel.cs
├── Views/
│   ├── Account/
│   ├── Dashboard/
│   ├── Grievance/
│   ├── User/
│   ├── Role/
│   ├── Department/
│   ├── Category/
│   └── Report/
├── Services/
│   ├── IGrievanceService.cs
│   ├── GrievanceService.cs
│   ├── IUserService.cs
│   ├── UserService.cs
│   └── IReportService.cs
├── Repositories/
│   ├── IRepository.cs
│   ├── Repository.cs
│   ├── IGrievanceRepository.cs
│   └── GrievanceRepository.cs
├── Data/
│   ├── ApplicationDbContext.cs
│   └── DbInitializer.cs
├── wwwroot/
│   ├── css/
│   ├── js/
│   └── images/
├── Helpers/
│   ├── ValidationHelper.cs
│   ├── ExceptionHandler.cs
│   └── AuthorizationHelper.cs
├── appsettings.json
├── Program.cs
└── Startup.cs
```



Professional Project Summary

The Grievance Management System is a robust and scalable ASP.NET Core MVC full-stack application developed to digitize and streamline the grievance handling process within an organization or institution. The system ensures proper registration, assignment, monitoring, escalation, and resolution of grievances through a secure and role-based platform.

KEY FEATURES AND CAPABILITIES:

1	Comprehensive Grievance Lifecycle Management - End-to-end workflow from submission to closure - Automated grievance number generation - Complete audit trail and history tracking - Status progression with clear accountability
2	Role-Based Access Control - Four primary user roles: Admin, Grievance Officer, Department Head, and User - Granular permission management - Secure authentication using ASP.NET Core Identity - Session management and logout functionality
3	Advanced Grievance Processing - Multi-level escalation support - Assignment and reassignment capabilities - Remarks and action tracking - Attachment management for supporting documents
4	Comprehensive Reporting and Analytics - Role-based dashboards with key metrics - Department-wise, category-wise, and priority-wise reports - Date-range filtering and advanced search - Export to Excel/PDF functionality
5	Data Integrity and Security - Normalized relational database design - Strong referential integrity with foreign keys - Soft delete mechanism for data preservation - Comprehensive audit logging
6	Professional User Interface - Responsive Bootstrap-based design - Intuitive navigation and user experience - Form validation and error handling - Mobile-friendly interface

TECHNICAL EXCELLENCE:

- Layered architecture with clear separation of concerns
- Repository pattern for data access abstraction
- Service layer for business logic encapsulation
- Dependency injection for loose coupling
- Entity Framework Core for ORM functionality
- SQL Server for robust data management
- Git/GitHub for version control

PROJECT SUITABILITY:

- Academic demonstration of full-stack web development
- Professional implementation in educational institutions
- Corporate grievance management systems
- Government and public sector organizations
- Any organization requiring structured grievance handling

DELIVERABLES:

- Fully functional ASP.NET Core MVC application
- Comprehensive database with 13 normalized tables
- 40+ screens covering all functional requirements
- Role-based dashboards and reporting modules
- Complete documentation and project structure
- Professional code organization and best practices

CONCLUSION:

The Grievance Management System represents a complete, production-ready solution for managing organizational grievances with transparency, accountability, and efficiency. With its structured database, clear workflow management, comprehensive reporting capabilities, and professional architecture, the project demonstrates advanced full-stack development skills and is suitable for both academic evaluation and real-world organizational deployment.